

13 Language Learning for Work Proficiency

SOCIAL

Awareness
Collaboration
Coordination
Articulation work
Conversation
Interaction
Mutual learning
Presence
Activity traces
Synchronous/Asynchronous

MOBILE

Context-specific
Context-diverse
Context-independent
Context-aware
Location-specific
Location-aware
Transience
Public/private
...

Social:

- Articulation work
- conversation
- interaction
- mutual learning
-

Mobile:

- context-specific
- context-aware
- location-aware

Target:

- Graduate, international students

Current problems:

- Most language learning apps are for individual learning, applying the language in a social setting is harder due to unfamiliarity, let alone professional social settings
- Language + social context = more effective learning?

- Learning in a group with a situation that pressures people to learn the language and interact with each other, like a collaborative situation (e.g. case study)

Context Ideas:

- Company on-boarding for newbies for one specific office and industry
 - asynchronous / synchronous???

Feedback

training human-interaction with computer-interaction

be aware of the role you put the technology as

alternatives

- the co-worker who wants to support the worker as a target, the robot introduces phrases and sentences to support the teaching
- theres a human coach but the robot support the coach

be aware of what the technology is adding or taking away

what does the technology do specifically?

- supporting who?
- teaching what?

Coworker as the target, the teacher is supported instead of the learner supported.

Robot is owned by the workplace

Research on language learning:

<https://dl.acm.org/doi/10.5555/3721488.3721612>

Summary

This is a replication study from HRI 2025 examining whether a humanoid robot-based language learning system (built on Self-Determination Theory — autonomy, competence, relatedness) works as well for students with dyslexia as it did for typical students in prior research.

Setup: 90 students with dyslexia (treatment group, learning with the robot) and 40 students with dyslexia (control group, learning with a teacher) went through matched Cantonese/Traditional Chinese learning materials over 5 days. Engagement (behavioural, emotional, cognitive) and intrinsic motivation were measured via pre/post questionnaires; learning outcomes via proficiency tests.

Key findings:

- The robot-taught (treatment) group showed significant gains in emotional engagement, cognitive engagement, and intrinsic motivation; behavioural engagement improved but wasn't statistically significant.
- The teacher-led (control) group showed no significant improvements, and in some measures (cognitive engagement, motivation) actually declined slightly.
- Treatment group proficiency improved by 17.26%, notably higher than the 9.88% improvement seen in typical students from the earlier study — suggesting students with dyslexia may benefit even more from robot-assisted learning.
- Limitations: small sample, single educational context, short duration.

Pain Points — What Language Learners (Especially Those with Dyslexia) Struggle With

Struggles inherent to dyslexia in language learning:

- Difficulty with reading and writing tasks generally, which underlies most classroom language instruction
- Greater susceptibility to demotivation from repetitive or monotonous tasks (standard drilling exercises backfire)
- Lower behavioural, emotional, and cognitive engagement compared to typical peers at baseline
- Passive reception of information in traditional teacher-led settings, which can suppress rather than build engagement

Struggles with traditional (teacher-led) instruction specifically:

- Motivation and engagement tend to *decline* over the course of instruction rather than build
- Lack of individualized pacing or feedback — one-size-fits-all delivery doesn't accommodate different processing needs
- Peer influence and social comparison in classroom settings can negatively affect confidence and participation (the study even physically separated desks to reduce this)
- Reduced sense of autonomy — students have little control over pace, repetition, or how material is presented, which the paper ties directly to lower intrinsic motivation

Struggles that interactive/robot-based systems specifically tried to address:

- Fear of failure or embarrassment when answering incorrectly (addressed via non-judgmental, consistent robot feedback regardless of right/wrong answers)

- Needing multiple attempts/repetition to consolidate memory (addressed via 3 trials + reinforcement)
- Losing engagement without emotional/social feedback loops (addressed via robot's gestures, cheering, songs)
- Needing scaffolded progression rather than jumping to complex material (addressed via vocabulary → sentence expansion structure)

The core narrative: dyslexic learners' struggles aren't primarily about *capability* but about **autonomy, pacing, and emotional safety** — and when those needs are met (as the robot system attempted), engagement and outcomes improve more dramatically than for typical students.

Concept: "Onboarding Buddy" — Robot Coach + Mobile Companion

Core idea: A robot lives at a home base (e.g., a "practice corner" or HR desk) for structured drills, and a mobile app extends that same AI persona into the employee's actual workday — nudging, listening, and supporting language use in real moments, not just rehearsed ones.

How it hits SOCIAL

Dimension	How it's designed in
Mutual learning	Robot adapts scripts based on what phrases/topics the employee struggles with; also learns the workplace's actual jargon over time from real interactions it observes (with consent)
Conversation	Roleplay practice at the robot, real conversation coaching via the app
Collaboration/Coordination	Robot pairs the employee with a human "language buddy" (a willing coworker) and schedules/coordinates short practice chats between them

Articulation work	App suggests phrasing in the moment before a meeting or interaction ("Here's how to ask this in a way that lands well")
Presence	App shows if the employee's language buddy or coach is currently available for a quick check-in
Activity traces	Manager/mentor can see a light progress trail (not surveillance — just "practiced 4x this week, confidence rising in small talk")
Synchronous/Asynchronous	Robot sessions are synchronous (scheduled); app nudges are asynchronous (available anytime, non-blocking)
Awareness	App gives light social awareness — "your team has a standup at 10am, here are 3 phrases you might need"

How it hits MOBILE

Dimension	How it's designed in
Location-aware / Location-specific	App detects if the person is near the kitchen, meeting room, or floor and adjusts — casual phrases near the kitchen, formal phrasing near the meeting room
Context-aware / Context-specific	Draws from calendar data — different prep for "1:1 with manager" vs. "team lunch"

Context-diverse	Same tool works across settings: onsite floor, remote calls, hallway chats — not locked to one room like the stationary robot alone
Transience	Nudges are short-lived, glanceable — a phrase suggestion that appears 5 minutes before a meeting and disappears after, not a persistent lesson module
Public/private	Practice with the robot is private (low-stakes, no judgment); real-time app nudges are private too (earpiece or silent notification, not visible to coworkers)
Context-independent	Baseline vocabulary drills (during robot sessions) work the same regardless of where the employee is that day

Why this works better than "robot alone"

The stationary robot by itself only covers **rehearsal** — it's social in a sandboxed way, and not mobile at all since it never leaves its spot. By splitting the system into **robot (deep practice, low-stakes) + app (live, contextual, in-the-moment support)**, you get:

- The *safety* of rehearsing with a non-judgmental robot (addresses the fear-of-failure problem from your earlier paper on dyslexic learners — same principle applies to language learners in a new workplace)
- The *realism* of support during actual interactions, which is where most language transfer struggles happen anyway (rehearsed fluency often collapses under real social pressure)

Where it genuinely enhances human interaction

1. Lowers the barrier to a first attempt

The biggest failure mode in workplace language struggles isn't ability — it's the fear of the *first* awkward interaction. Practicing with the robot first (low-stakes, no social cost to messing up) means the employee's first real attempt with a human coworker is their *second* attempt,

not their first. That changes the quality of the actual human exchange — less anxiety, more presence.

2. Creates a legitimate reason for coworkers to connect

The "language buddy" pairing isn't just translation help — it's a structured excuse for two people who might never otherwise talk to have a recurring, low-pressure conversation. That's a real social bond, not a simulated one. The tool is a matchmaker/scaffold, but the actual interaction is 100% human.

3. Improves the *quality* of interactions that would've happened anyway

The in-the-moment nudges (before a standup, before a 1:1) mean the employee shows up more prepared to actually engage — asking a real question instead of nodding along, using the right register instead of a stiff or overly formal phrase. That makes the human conversation richer, not replaced.

4. Reduces one-sided burden

Normally, the coworkers around a new hire absorb the "teaching" cost informally, often awkwardly. If the robot/app absorbs some of that groundwork, human interactions can be more equal — less "coworker patiently explaining basics," more "two people actually talking."

Where it risks *not* enhancing it (worth designing around)

- **Over-reliance risk:** if the app becomes a crutch the employee leans on *during* every real conversation (checking phrases mid-conversation), it could make interactions feel more mediated and less present — the opposite of the goal. The nudges should happen *before*, not *during*, live conversation.
- **Rehearsed feels rehearsed:** if practiced phrases sound too scripted, coworkers may sense it and the interaction feels less genuine. The system should coach toward natural variation, not fixed lines.
- **Risk of substituting for human contact:** if rehearsal with the robot feels *safer and easier* than real coworkers, some people may over-invest in it and under-invest in real connection. The buddy-pairing feature is partly a safeguard against this — the loop should always terminate in a human, not stay contained in the app.

Bottom line: the system enhances human interaction mainly by acting as *preparation and confidence infrastructure*, not a participant in the interaction itself. Its value is in what it does *before and around* the human conversation, not inside it — the moment it starts doing the talking *for* the person, it stops enhancing social interaction and starts replacing it.

Owned by a company. Located only in the workspace.

[https://www.theseus.fi/bitstream/handle/10024/899433/Onboarding Practices for Cross.pdf?sequence=1](https://www.theseus.fi/bitstream/handle/10024/899433/Onboarding_Practices_for_Cross.pdf?sequence=1)

Suggested recommendations have mentioned below.

1. Implement cultural sensitivity training

- a. Provide cultural awareness training for all employees, addressing cultural norms, communication styles, and workplace behaviour. This will improve understanding, reduce miscommunication, and foster a respectful workplace.
2. Establish a mentorship program
 - a. Connect new hires with experienced colleagues who understand diverse cultures. These mentors will offer tailored support throughout the first year, helping newcomers understand cultural subtleties, smoothly adapt to the workplace, and feel valued.
3. Introduce a buddy system
 - a. Match new employees with a "buddy" who shares a similar cultural background. This peer will provide guidance and assistance during the initial year, easing the transition, boosting confidence, and fostering a sense of belonging.
4. Incorporate feedback mechanisms
 - a. Employ regular feedback mechanisms, such as anonymous surveys, to evaluate the experiences of new employees at various stages (e.g., one week, one month, and three months). These tools help pinpoint ongoing issues and identify areas for enhancement.
5. Foster an inclusive company culture
 - a. Promote open dialogue and respect among employees through activities like team-building events, diversity training, and inclusive company guidelines. A welcoming environment increases employee happiness, involvement, and loyalty.
6. Enhance team integration activities
 - a. Plan team-building activities and cultural exchange programs, both online and in person, to help employees connect and build understanding. These activities will help overcome cultural differences and strengthen relationships within teams.
7. Monitor and evaluate onboarding practices
 - a. Regularly assess and refine the onboarding process to ensure it meets the needs of the workforce. This allows for ongoing improvements and adjustments to address any weaknesses.
8. Create a diverse onboarding team
 - a. Create an onboarding team composed of individuals with various cultural backgrounds to develop inclusive practices and provide support to new employees during their initial period.

Following these suggestions will improve Company X's onboarding, leading to better integration and happier, more productive diverse employees.

7. Conclusion

This research investigates how to improve the onboarding process for new employees at Company X, specifically focusing on those from diverse cultural backgrounds. The current standard process is not effective and causing problems. The study emphasised the importance of considering cultural differences during onboarding and suggested changes to create a more welcoming and inclusive experience. The potential benefits include happier and more productive employees who integrate well into the company's culture. The research recommends ongoing feedback to continually improve the onboarding.

Although this study was constrained by a limited number of participants and a specific scope, future research could broaden its reach by involving a larger and more diverse group of individuals across multiple locations. Additionally, exploring the long-term impacts of culturally tailored onboarding practices could provide deeper insights and further validate the findings. Ultimately, global companies should prioritise culturally sensitive onboarding to value diversity and treat their employees with respect.

Key pain points of a language learner

Pain point	What the learner experiences	Why it is a problem
Difficulty measuring progress	Learners may not know whether their language ability is actually improving.	Progress can feel unclear or slow, reducing motivation.
Language proficiency is multidimensional	Being good at one area doesn't necessarily mean being good at another.	A learner may understand grammar but struggle with actual communication.
Difficulty transferring knowledge into communication	Learners can know vocabulary/grammar but struggle to use it in real situations.	Classroom knowledge doesn't always translate into practical language use.
Difficulty understanding language in context	Knowing individual words isn't enough to understand meaning in real communication.	Context, interpretation and appropriate language use affect communication.
Speaking/communication challenges	Learners may struggle to produce language effectively despite understanding it.	This can make real-world interactions stressful or frustrating.
Testing does not capture everything	A test score may not fully represent someone's ability to communicate.	Learners can feel that their score doesn't reflect their actual skills.
Different skill levels across areas	Reading, writing, listening and speaking can develop at different rates.	A learner may feel confident in one skill but weak in another.

Lack of meaningful feedback	Learners may receive a score/grade without understanding exactly what they need to improve.	It becomes difficult to decide what to practise next.
Difficulty knowing what to learn next	Learners can be overwhelmed by vocabulary, grammar, pronunciation and other areas.	Lack of direction can make learning inefficient.
Maintaining motivation	Language learning requires sustained effort over time.	Slow or unclear progress can cause learners to lose motivation

<https://journals.sagepub.com/doi/abs/10.1177/0002716287490001003>

Tutor feedback

- Keep searching the acm library
- Socioconstructivism
- Keep it broad
- Include the iteration of fashion → language
- Can include the potential solution as a concept
- Iterations and elaborations of the initial idea
- Propose different solutions
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